



MSc Research Ethics Committee



MSc Student
Medical Statistics
LSHTM

25 April 2026

Dear

Study Title: Examining the Timeliness of Routine Vaccination During Heatwaves

LSHTM MSc Ethics Ref: 33670

Thank you for your application for the above MSc research project, which has now been considered by the MSc Research Ethics Committee.

Confirmation of ethical opinion

On behalf of the Committee, I am pleased to confirm a favourable ethical opinion for the above research on the basis described in the application (CARE) form, and supporting documentation, subject to the conditions specified below.

Conditions of the favourable opinion

Approval is contingent on local ethical approval having been received, where relevant. It is the responsibility of the student and their supervisor to ensure appropriate local ethical approval is in place before a study commences (ie if you indicated in the local approval section that local approval is required).

Approved documents

The final list of documents reviewed and approved by the Committee is as follows:

Document Type	File Name	Date	Version
Other	Screenshot 2026-03-25 at 21.02.46.png	25/03/2026	v1
Other	Screenshot 2026-03-25 at 21.45.23.png	25/03/2026	gps
Other	AuthLetter_230640.pdf	26/03/2026	granted-access
Investigator CV	pdf	17/03/2026	v1

After ethical review

Any subsequent changes to the application **must** be submitted to the Committee via an Amendment form on the ethics online applications website: <http://leo.lshtm.ac.uk>. Amendments must not be initiated before receipt of written favourable opinion from the committee.

Yours sincerely,

Professor Paul Milligan
Chair

MScEthics@lshtm.ac.uk
<http://www.lshtm.ac.uk/ethics>

Improving health worldwide



Mar 25, 2026



Request Date: 03/25/2026

Dear 

This is to confirm that you are approved to use the following Survey Datasets for your registered research project titled: "Examining the Timeliness of Routine Vaccination During Heatwaves":

Nigeria

To access the datasets, please login at: https://www.dhsprogram.com/data/dataset_admin/login_main.cfm. The user name is the registered email address, and the password is the one selected during registration.

The IRB-approved procedures for DHS public-use datasets do not in any way allow respondents, households, or sample communities to be identified. There are no names of individuals or household addresses in the data files. The geographic identifiers only go down to the regional level (where regions are typically very large geographical areas encompassing several states/provinces). Each enumeration area (Primary Sampling Unit) has a PSU number in the data file, but the PSU numbers do not have any labels to indicate their names or locations. In surveys that collect GIS coordinates in the field, the coordinates are only for the enumeration area (EA) as a whole, and not for individual households, and the measured coordinates are randomly displaced within a large geographic area so that specific enumeration areas cannot be identified.

The DHS Data may be used only for the purpose of statistical reporting and analysis, and only for your registered research. To use the data for another purpose, a new research project must be registered. All DHS data should be treated as confidential, and no effort should be made to identify any household or individual respondent interviewed in the survey. Also, be aware that re-distribution of any DHS micro-level data, either directly or within any tool/dashboard, is not permitted. Please reference the complete terms of use at: <https://dhsprogram.com/Data/terms-of-use.cfm>.

The data must not be passed on to other researchers without the written consent of DHS. However, if you have coresearchers registered in your account for this research paper, you are authorized to share the data with them. All data users are required to submit an electronic copy (pdf) of any reports/publications resulting from using the DHS data files to: references@dhsprogram.com.

Sincerely,

Olsen Hanner

Olsen Hanner
Data Archivist
The Demographic and Health Surveys (DHS) Program

Ethics Application & CARE Form

Project Information

Staff members/students based at:

- ☒ LSHTM
- ☐ MRC Gambia@LSHTM (For LSHTM staff/students requiring SCC/ GG/MRCG EC review, please select this option)
- ☐ MRC Uganda@LSHTM
- ☐ Other external applicant applying to the SCC/ GG/MRCG EC only

1. Full project title

Examining the Timeliness of Routine Vaccination During Heatwaves

1a. Basic project information will be made available on the LSHTM website following a favourable ethical opinion. Do you require your project information to be withheld? (see information icon for details on the information that will be made available) (this does not apply to MSc student projects).

- ☐ No
- ☐ Yes
- ☒ Not applicable (MSc student)

2. Is this Project in fulfillment of a degree?

- ☒ Yes
- ☐ No

2a. Degree registered for

MSc

2e. Indicate student type

- ☐ Distance Learning
- ☒ Face-to-Face or Intensive Online

2f(MSc) Is this an original submission, or are you responding to a request for clarification or insufficient information notification from the LSHTM Ethics Committee?

- ☒ Original submission
- ☐ Responding to the Ethics Committee
- ☐ Resubmission following an unfavourable opinion

Student Details

3a. Student details

Title

First Name

Surname

[Redacted]

[Redacted]

[Redacted]

[Redacted]

Telephone

[Redacted]

Email

[Redacted]

[Redacted]

[Redacted]

3a(i). Please provide your student ID number

[Redacted]

3a(ii). MSc course

Medical Statistics



3a(iii). Indicate proposed project length:

- ☒ Standard
- ☐ Extended

3b. Are you studying as a full or part time student?

Full time



3c. Supervisor's name.

[Redacted]

3c Supervisor's email address (if more than one, please only provide the email address of your main supervisor)
(i).

Email

3 Supervisor's institution
c(ii).

- ☒ LSHTM
☐ MRC Gambia or Uganda
☐ Other

3e. Supervisor status

Confirmed

Project Type

Note: Completing the filter will enable and disable sections of the form so you may not see all questions.

4. Does the research involve primary data collection, analysis of data/samples that have already been collected, or a mix of both?

- ☐ Primary
☒ Previously collected data/samples
☐ Mixed

4a(iii). Select type of project:

Project using data from secondary sources

4c. Does the project involve extraction of data from patient records (e.g. medical, social care, service user records)? (This refers to primary data collection from records and does not include data that was previously collected and is now being used in a secondary analysis).

- ☐ Yes
☒ No

6. Does this project require review by the Commercialisation and Rapid Response (CaRR) ethics committee? (please see info icon for the remit of this committee)

- ☐ Yes
- ☒ No

Samples

6a. Does this research project involve the collection, or use of previously collected, human biological specimens e.g urine, stool, blood etc? (Please select yes even if the samples are not considered relevant material under the Human Tissue Act)

- ☐ Yes
- ☒ No

6b. Will this project involve living animals (either laboratory, livestock or wild animals) AND/OR biological material that has been obtained from animals in the experiments planned?

- ☐ Yes
- ☒ No

Fast-Track

7. Does this project use anonymised and unlinkable secondary datasets only? (If yes the application may be eligible for fast-track 7b.)

- ☒ Yes
- ☐ No

7a. Will this project be conducted in conjunction with NHS staff, premises or any other connection to the NHS?

- ☐ Yes
- ☒ No

7b. Is this application for fast-track? Please see information icon for which projects are eligible. Note: MSc applications are not eligible for fast-track.

- ☐ Yes
- ☒ No

Study Population and Vulnerable Groups

Please answer the following questions with respect to the research subjects / participants who you are seeking ethical approval to research in this application

8c. Please select the age ranges of the participants involved in the original study. Select all that apply.

- ☒ infants/children (<10 years)
- ☐ Children and adolescents (10-17 years)
- ☐ Adults (18+ years)

Security Sensitive Research Material

9. Does this research involve access to and/or storage of security sensitive research material? Please note that while some data is considered sensitive, such as HIV status, it is not necessarily considered security sensitive. If you are using data that could be considered sensitive, but not security sensitive please answer no to whether your research involves access to and/or storage of security sensitive research material. Please see information icon for what is considered security sensitive material.

- ☐ Yes
- ☒ No

Geography

10. List the countries where the research project is to be conducted (For example: if you are conducting a secondary data analysis for your project and you will be based in the UK, select UK regardless of where the original data has come from):

United Kingdom

10a. For projects using secondary data, please list the countries where the original data has come from.

Nigeria

10c. Does this project involve primary health research in the UK?

- ☐ Yes
- ☒ No

Outline

Note:

When using an acronym, spell it out in full the first time it is mentioned, with the acronym following in round brackets.

Please do not copy and paste directly from the protocol. Applications where large portions of text have been copied and pasted directly from the protocol, and therefore do not properly answer the question, will be invalidated

11. Are you seeking ethical approval for one or more components of a multi-component/stage/phase study?

☐ Yes

☒ No

11c. Proposed start date of the project

01/05/2026

The date provided here should be the proposed start date of the project (when you plan to start your research involving humans/human data/human biological specimens). This should be after all required approvals are in place. If this project is part of a larger study, please give the proposed start date for this project, not the start date of the larger study. For MSc students, please give the date you plan to start your project, not the start date of the module. Please note, the ethics committee cannot provide retrospective review.

11d. Proposed end date of the project

02/09/2026

This is the proposed end date of your project. This should be after your proposed start date. End of study should be defined by the PI. If your project continues beyond the date given in this application you are required to submit an amendment to extend the end date.

12. Please give an executive summary of the proposed project, including background, methods, and aims of the proposal. Include information from any systematic reviews that have been conducted. Sufficient detail must be given to allow the Committee to have a good overview of the project without reference to other documents. The summary should be written in such a way that is suitable for lay members of the committee.

Aim:

This project aims to investigate whether children whose routine vaccinations are due during periods coinciding with heatwaves are more likely to experience delays compared with children scheduled during periods without heatwaves.

Background:

Routine childhood vaccination is one of the most effective public health interventions for preventing infectious disease and reducing child mortality. However, the benefits of vaccination depend not only on whether children are vaccinated, but also on whether they receive vaccines at the recommended time. Delays in routine immunisation leave children vulnerable to preventable diseases and can contribute to sporadic outbreaks. At the same time, climate change is increasing the frequency and severity of extreme weather events including heatwaves. Heatwaves can affect health systems and community behaviour in several ways. Importantly, extreme heat may make it difficult for caregivers to travel to health facilities, reduce attendance at vaccination clinics, or disrupt health services. Despite these plausible pathway, there is limited evidence on whether heatwaves influence the timeliness of routine childhood vaccination. Systematic reviews of vaccination timeliness have identified several important determinants, including maternal education, socioeconomic status, rural residence, and distance to health facilities. These studies highlight that structural barriers and access to care are key drivers of delayed vaccination. However, environmental factors such as extreme weather events have not been studied in relation to vaccination timeliness.

Methods:

The study will focus on Nigeria, using survey data from the 2024 Demographic and Health Survey (DHS). These data include information on children's vaccination histories and household characteristics. To identify periods of extreme heat, the study will use meteorological data derived from satellite-based products, specifically ERA5 (European Centre for Medium-Range Weather Forecasts Reanalysis Version 5). These datasets provide detailed temperature information across time and location. Heatwave "windows" will be defined according to the World Weather Attribution (WWA) protocol. These heatwave records will be linked to the survey data using information on the timing and approximate location of each child's vaccination schedule. The main analysis will then examine whether the probability of delayed vaccination is higher when a child's scheduled vaccination date falls within a heatwave window, taking into account demographic and other social factors of the child.

12b. Upload the study protocol (compulsory for staff and doctoral students), including data collection forms, questionnaires and topic guides. Please upload each document separately, ensuring that the date and version number of each document is correct.

13. State the intended value of the project, detailing why the topic is of interest or relevance. If this project or a similar one has been done before what is the value of repeating it? Give details of overviews and/or information on the Cochrane database. This area is of increasing importance – please ensure you give a full response.

Heatwaves have been associated with increased mortality and increased morbidity in multiple geographic contexts. The investigation into its effect on vaccine timeliness is a meaningful extension, that allows us to get a better understanding into how extreme hat events increase the burden on public health by delaying vaccine uptake, challenging vaccine supply chains. Similar work has been done in countries such as Pakistan, however, this work remains geographic dependent, as the definition of a heatwave itself changes from country to country, and its association with vaccine timeliness will ultimately need to take into account the demography and social context of countries under investigation.

13a. Please outline what benefits are anticipated for those people taking part, the community/population the study is conducted in, and the broader population.

1. The study seeks to inform relevant bodies such as the ministry of health on the impact of heatwaves on vaccine uptake. This allows the relevant bodies to plan in advance according to weather forecasts or seasonal weather outlooks, build resilient vaccine infrastructure and conduct campaigns such as door-or-door campaigns in areas most affected by heatwaves.
2. The estimates from the study, should they point to heatwaves being risk factors allow public health bodies to estimate the potential outbreak sizes that might occur, due to children missing vaccine timelines.
3. This study will inform the government and ministry of health on the logistics for conducting catch-up campaigns and other supplementary immunisation campaigns in areas most affected by vaccine delays.

13b. Do you believe the study poses any particular ethical challenges that you would like to draw to the attention of the ethics committee? If yes please outline the challenges.

The study does not pose any ethical challenges.

14. Hypothesis statement.

Heatwaves are associated with delays in childhood vaccine uptake.

15. Overall aim of project

This project aims to investigate whether children whose routine vaccinations are due during periods with heatwaves are more likely to experience delays compared with children scheduled during periods without heatwaves.

16. Specific objectives of project

1. To determine the average duration of time-to-vaccination and timeliness probabilities for each dose in Nigeria's childhood schedule.
2. To quantify the association between heatwave exposure at the scheduled vaccination date and the likelihood of dose-specific delay (we use two thresholds: defined as >2 weeks past recommended time as well as >4 weeks past the recommended time) taking into account socio-demographic factors.

Note: Please do not copy and paste directly from the protocol. Applications where large portions of text have been copied and pasted directly from the protocol, and therefore do not properly answer the question, will be invalidated

18. Specify the procedures/methodology to be conducted during the project. Please include outcome measures and plans for data management and analysis. For literature reviews, include details on search strategy, search terms, inclusion and exclusion criteria.

Data

The data will be requested from the DHS server for Nigeria's 2024 DHS survey, together with its geo-coded file, for spatial analysis and data visualisation. Analysis will follow the DHS 8 guide for analysing DHS data in accordance with the DHS guideline. Known confounders will be obtained from a systematic review conducted on vaccine delay and timeliness in Low and Middle Income Countries, and inclusion of variables will be subject to availability in the DHS dataset itself.

Outcome measures

The time to vaccination is calculated by taking the difference between the date of expected vaccination for each antigen and the reported date for vaccination. The expected date of vaccination is calculated from the reported date of birth for each child, and the date the child will reach the expected age based on the vaccination calendar for Nigeria. While Nigeria has 24 antigens within its routine immunisation programs, DTP, PCV, MCV are delivered through routine services and these will be the main outcomes that we will explore. The remaining vaccines could be explored as a secondary objective. The unit of analysis will be the individual children, and the analysis will take into account the clustering in the multi-stage stratified cluster sampling design.

Data analysis plan

The time-to-vaccination (outcome variable) will have a distribution, which we expect to be approximately normal, but as part of the data analysis we will explore what distribution best describes (eg. Weibull) the variability in the outcome variable. We will explore trends in vaccination timeliness, for example across regions in Nigeria, and the consistency of timeliness across antigens. A generalised linear regression model will be used to explore variables associated with an increase in the outcome, specifically including the presence of heatwaves within the region of residence at the time of expected vaccination. All model estimates and generated quantities will be summarised using the median and their 95% credible intervals.

23. In terms of the feasibility of your project, what could stop this project from succeeding, or prevent you from achieving your objectives? *Please indicate any aspects of your proposed approach which could potentially experience difficulties, e.g. delays with permissions, data collection or storage problems, lack of sufficient comparable information, etc. You may also wish to mention any wider matters which could affect your project, e.g. civil unrest, natural disasters, transport availability.

While there exists limited formal epidemiological evidence, anecdotal reports and local observations suggest that extreme heat events and heatwaves have occurred in several regions in Nigeria, however it may be possible that no heatwave windows coincided with some child records during the period when the Nigeria DHS 2024 was being conducted.

24. What alternative plans do you have in case you encounter any of the potential problems you have identified? Please troubleshoot all points raised in question 23.

1. We may use earlier surveys such as Nigeria 2008, 2013, 2018 DHS surveys.

25. What specific facilities or resources will you personally expect to make use of for your project (eg a local university library, lab facilities, project placement with a specific organisation etc)?

None.

27. List key references (for MSc students no more than 5, for staff and doctoral students please include a minimum of 10), including for methods to be used.

Imran H, Raja D, Grassly NC, Wadood MZ, Safdar RM, O'Reilly KM. Routine immunization in Pakistan: comparison of multiple data sources and identification of factors associated with vaccination. *Int Health*. 2018 Mar 1;10(2):84-91. doi: 10.1093/inthealth/ihx067. PMID: 29432552; PMCID: PMC5856161.

Siddiqi, D. A., Iftikhar, S., Anfossi, C. M., Siddique, M., Shah, M. T., Dharma, V. K., ... & Chandir, S. (2025). Assessing the impact of heat waves on childhood immunization coverage in Sindh, Pakistan: Insights from 132.4 million doses recorded in the provincial electronic immunization registry (2018–2024). *Vaccine*, 61, 127424.

Akatak, B. M., Onyeaka, H., Onungwe, I., Akpan, P. P., Igulu, K. T., & Elenwa, E. P. (2025). Climate change-induced heatwaves in Nigeria: Causes, challenges, and adaptive strategies. *Journal of Environmental Management*, 394, 127433.

Wagner, A. L., Tefera, Y. A., Gillespie, B. W., Carlson, B. F., & Boulton, M. L. (2021). Vaccine coverage, timeliness and delay estimated from regional and national cross-sectional surveys in Ethiopia, 2016. *The Pan African Medical Journal*, 39, 205.

Experience

30. State the personal experience of the applicant and of senior collaborators in the research project in the field concerned, and their contribution to this project. Indicate any previous work done related to the project topic including student and/or professional work, or publications

I have previously worked in a research setting as a research assistant, where we utilised the DHS data extensively together with Earth Observation data from satellites to answer public health and epidemiology questions. My experience allows me to tackle the questions being proposed on this project.

- 30a. Upload the CVs for all main investigators working on the project. For MSc students, please upload your CV only.

- 30e. Have the main investigators (all those named in 3f.) undertaken any Research Ethics/Human Subjects Protection training? It is recommended that all applicants and members of their research team should complete ethics training at least every 3 years to account for changes in guidelines and regulations. Links to online training can be found in the information icon. (Please note this is not the same as GCP training).

☐ Yes

☒ No

- 30e(ii). If the main investigators have not undertaken any Research Ethics/Human Subjects Protection training please explain why.

I am taking it, just have not completed it yet.

Informed Consent - Secondary Data

34. Is participant consent in place for secondary use of the data?

- ☒ Yes
- ☐ No

34c. Please give details of the participant consent that was obtained when the original project(s) took place. Please upload copies of the original information sheet and consent form(s) (Please provide English translations where relevant). If there are no original informed consent forms (e.g. for audit or DHS data) please explain this.

I am planning to use the DHS data, specifically information on the birth recode dataset.

They have this information on their webpage: <https://dhsprogram.com/methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm>

Ethical Review

Procedures and questionnaires for standard DHS surveys have been reviewed and approved by ICF Institutional Review Board (IRB). Additionally, country-specific DHS survey protocols are reviewed by the ICF IRB and typically by an IRB in the host country. ICF IRB ensures that the survey complies with the U.S. Department of Health and Human Services regulations for the protection of human subjects (45 CFR 46), while the host country IRB ensures that the survey complies with laws and norms of the nation.

For the ERA5 Data from Climate Data Store, the data is publicly available for download.

Confidentiality & Data

40. State how your data will be stored and what will be done with it at the end of the project (please see information icon for guidance).

Data will be stored on my computer, during the period I will be conducting the project, as I will be analyzing it strictly from my computer. Once the project is done, submitted and accepted. I will destroy the dataset by securely deleting it, in line with DHS rules.

Funding

46. Do you have external funding for this project?

- ☐ Yes
- ☒ No

46b. How will the project go ahead without funds?

The project is using secondary data, as such, it will not need any funding.

46c. Are you in receipt of any funding from the United States? Or will you be collaborating with (or with individuals from) a US Institution/organisation?

- ☐ Yes
- ☒ No

47. Has the project been sent out for peer/independent scientific review (please select yes if the project is being sent to the SCC)?

- ☐ Yes
- ☒ No

47a. If no, why has the project not been sent peer/independent scientific review?

The project will be reviewed by my supervisor.

49. Does the Chief Investigator or any other investigator/collaborator have any direct personal involvement (e.g. financial, share holding, personal relationship etc.) in the organisations sponsoring or funding the research that may give rise to a possible conflict of interest?

- ☐ Yes
- ☒ No

50. Will individual researchers receive any personal payment over and above normal salary, or any other benefits or incentives, for taking part in this research?

- ☐ Yes
- ☒ No

Local Approval

69. For projects using previously-collected human data, give details of all approvals under which the original project(s) took place. Please quote names of Ethics Committees and approval reference numbers (required even if previous approval was from LSHTM); if possible give web link to original project application. If there are no original approvals (e.g. for audit or DHS data) please explain this.

From the DHS website: <https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm>
Procedures and questionnaires for standard DHS surveys have been reviewed and approved by ICF Institutional Review Board (IRB). Additionally, country-specific DHS survey protocols are reviewed by the ICF IRB and typically by an IRB in the host country. ICF IRB ensures that the survey complies with the U.S. Department of Health and Human Services regulations for the protection of human subjects (45 CFR 46), while the host country IRB ensures that the survey complies with laws and norms of the nation.

From the 2024 Nigeria DHS Final report:
The survey methodology and instruments received clearance from both the ICF Institutional Review Board ethics committee and the National Health Research Ethics Committee of Nigeria

69a. Will your analyses be for purposes entirely covered by the original ethics application where the data was collected, as detailed above?

- ☒ Yes, this falls within the aims and scope of the original project
- ☐ No, the analyses and aims differ from the original project

69a(i). Summary of purpose and methods of the original project(s)

The purpose is to look into how heatwaves affect the timeliness of vaccine uptake for selected childhood antigens in Nigeria. The methods will involve fitting regression models to the data, and generating quantities from the fitted model.

69b. Indicate how you will obtain permission to use the data from the principal investigator responsible for each original project

The DHS has a data request form, where you fill in your project title, and purpose, and select all datasets that you are requesting, and once approved, you will receive the permission and ability to download the datasets and use.
The climate data only needs you to login to the site and access the data, without any approvals, or permissions.

Data Sources, Results & Permissions

70. If you expect to use existing data, how will you obtain it? *Indicate who holds the data, who specifically you will contact, and by when. Any contact so far, especially anything confirmed in writing, should be mentioned.

I am planning to use climate data from Climate Data Store (CDS: (<https://cds.climate.copernicus.eu/>)), and the Demographic health Survey (DHS). For the CDS data, you only need to create an account, and you can freely download the data right away. With the DHS data, I need to submit a request to formally download the dataset, and upon review by the DHS team, they will give me access to download the data. I have already submitted the request for the DHS, and have been given permission to download.

70a. Please upload any documents confirming that you have requested/been granted permission to use any existing data. This can include email correspondence as well as formal letters of permission. (This is mandatory for projects using existing data).

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Other	Screenshot 2026-03-25 at 21.02.46	Screenshot 2026-03-25 at 21.02.46.png	25/03/2026	v1	839.7 KB
Other	Screenshot 2026-03-25 at 21.45.23	Screenshot 2026-03-25 at 21.45.23.png	25/03/2026	gps	844.3 KB
Other	AuthLetter_230640	AuthLetter_230640.pdf	26/03/2026	granted-access	31.7 KB

71. If you expect to use any public domain data, please give further details. *Make clear how you will gain access. Public domain data must be freely available to any member of the public, without any restrictions or requirement for special permission, and must not enable the identification of living people.

The datasets I am planning to use in the analysis are two. The climate data is freely available on the Climate Data Store upon creating an account (<https://cds.climate.copernicus.eu/>).
The next dataset I need is the DHS data which is also freely available upon creating an account, and requesting for it. I have already applied to obtain the Nigeria DHS 2024, and have been approved/given permission to download.

71a. Please provide the links to all public domain data that you expect to use.

<https://cds.climate.copernicus.eu/>
<https://dhsprogram.com/>

72. Will any specific data rights permissions or usage limitations be required regarding data to be used or collected in the project?

- ☐ Yes
☒ No

73. Will any agreements be required regarding data to be used or collected in the project (for example, material transfer agreements or data transfer agreements)?

- ☐ Yes
☒ No

74. Are there any existing obligations regarding ownership of results to third parties (e.g. employer)?

- ☐ Yes
☒ No

Type of risk

Students planning to travel as part of their project are strongly encouraged to get in touch with the Offsite Work Safety Officer at the earliest opportunity to discuss the requirements of their travel (offsite-safety@lshtm.ac.uk)

75. Will the project involve travel at any point, either within the UK or your primary country of residence, or outside of the UK/your primary country of residence (e.g. travel overseas)? Please note that this includes travel for any form of work for your project, not just primary data collection.

- ☐ Yes
☒ No

76. Please indicate which locations you expect to work at. Check all that apply.

- | | |
|---|--|
| ☐ Field Work | ☐ Lab Work |
| ☒ Desk-based (Home, LSHTM, Office, Library) | ☐ Other work away from home |

77. Will the project involve working with or handling any of the following materials?

- ☐ Pathogenic microorganisms (e.g. virus, bacteria, parasite)
☐ Human biological specimens
☐ Animal tissue
☐ Radiochemicals
☐ Genetically modified organisms
☐ Toxic chemicals
☒ None of the above

78. Are any other potentially hazardous activities likely to be carried out during the project?

- ☐ Yes
- ☒ No

Well-being

111. If you have a disability, mental health problem, or any other condition that may require support (e.g. colour blindness) and you will be working in a lab), it is strongly recommended that you contact Studentadvice@lshtm.ac.uk or Safety@lshtm.ac.uk in good time to discuss support options.

☒ I have read and understood the above information

112. Your wellbeing is of paramount importance and if your project research will mean spending time away from usual support networks, and if you think that you could find this challenging, we strongly encourage you to seek support and advice well in advance. You can book a confidential appointment with one of the Student Counsellors to discuss any concerns. The Counselling Service is available to all students and you do not have to have a diagnosed condition to seek support.

☒ I have read and understood the above information

Signature Instructions

The form should be completed and finalised prior to signing or requesting signatures. Incomplete or incorrectly filled in forms will be invalidated and this may cause a delay with your review. For external applicants/supervisors, please ensure that they have registered for an account prior to requesting the signature.

Please ensure that you obtain all signatures on this page. Failure to obtain all required signatures will result in the form not submitting and a delay to the processing of your application.

If you have trouble signing/requesting signatures please contact:

LSHTM London: ethics@lshtm.ac.uk or MScethics@lshtm.ac.uk

MRC Unit The Gambia at LSHTM: ethics@mrc.gm

For MSc students:

- Please indicate that you have read and understood each statement under the student declaration before adding your signature using the blue sign button at the end of the student declaration.
- Once you have signed, please request your supervisor signature using the request signature button under the supervisor signature section. For external supervisors, please ensure that they have registered for an account prior to requesting the signature.
- Once your Supervisor has signed, please request your Project Module Organiser signature using the request signature button under the Course Director / Project Module Organiser Signature section.
- Check whether any further signatures are required (these will show under the Project Module Organiser Signature section) and request as required.
- **Unless you have been advised otherwise, please submit signature requests in order (e.g. supervisor, followed by Project Module Organiser etc.), rather than all at the same time.**
- **Please ensure you have obtained all required signatures as your form will not submit without these. If you have any trouble please contact MScethics@lshtm.ac.uk**

Signature - Applicant

Student
Declaration

- ☒ I have read and understood, and agree to abide by the LSHTM Good Research Practice policy as well as all applicable Standard Operating Procedures, including on informed consent
 - ☒ I undertake to abide by all regulations, guidelines and standards of good practice, including but not limited to the Data Protection Act 2018, GDPR, and the Declaration of Helsinki
 - ☒ I undertake to abide by all local rules/laws for non-UK research
 - ☒ I agree to conduct my project on the basis set out in this form, and to consult staff (initially, my Supervisor) if making any subsequent changes
 - ☒ I agree to inform the ethics committee of any changes made to this form, and will not implement any changes until approval from the ethics committee has been received
 - ☒ I undertake to adhere to all conditions set out by review bodies in giving approval and will not start the project until all required approvals are in place
 - ☒ I agree to comply with the relevant safety requirements, and will submit a separate request for LSHTM travel insurance where relevant
 - ☒ I agree to inform the Faculty Safety Officer and/or the Off-Site Safety Advisor (as required) if there are any changes to the risk assessment
 - ☒ I confirm that there are no conflicts of interest that preclude my participation in the project
- Student signature
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Signature - Supervisor

Supervisor signature

I declare that:

- I agree that the information submitted in this application is a reasonable summary of the proposed project.
- I agree that this form correctly indicates whether or not ethics approval will be required.
- I agree that this form contains adequate information for the ethics committee to form an opinion of the proposed project.
- I agree that all required supporting documentation is attached to this application.
- (For MSc projects only) I agree that responses in the Risk Assessment section address the main risks connected with a project of this nature
- I have reviewed the risk of the project, including travel, and agree that it is an acceptable risk to the student
- I confirm that there are no conflicts of interest that preclude my role as supervisor for this project
- I Have read and understood, and agree to abide by the LSHTM Good Research Practice policy



Signature - Course Director/Project Module Organiser

Course Director / Project Module Organiser Signature

I declare that:

- I agree that the proposed project's academic content is suitable for this MSc

Signed: This form was signed by Dr. Melissa Neuman (Melissa.Neuman@lshtm.ac.uk) on 20/04/2026 12:05 PM

Signature - Other

Note:

The form will automatically submit upon receipt of all required signatures.

After submission, you will receive a confirmation email with further details.

For staff based at LSHTM London, if you have not received a confirmation email within 5 working days please email ethics@lshtm.ac.uk (staff) or MScethics@lshtm.ac.uk (students) to check the status of your submission.

For staff based at MRC Unit The Gambia at LSHTM, or external applicants, please contact ethics@mrc.gm